Task Two: California Housing Price Regression Model

Datheon Machine Learning & Data Science Internship | Technical Report

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1. Executive Summary

This technical report details the end-to-end development, exploratory data analysis, pipeline engineering, cross-validated hyperparameter tuning, and performance evaluation for Task Two of the Datheon ML & DS Internship. The goal was to build a robust regression pipeline to predict continuous California housing values (median_house_value), handle missing values via imputation, engineer scaled numerical and one-hot encoded categorical features, apply regularized linear models (Lasso & Ridge) alongside GridSearch cross-validation, and benchmark regression metrics (RMSE, MAE, R^2).

> **TASK BENCHMARK & MODEL PERFORMANCE**
> The fine-tuned Ridge Regression model achieved the optimal test set performance with a Root Mean Squared Error (RMSE) of $69,150.31, Mean Absolute Error (MAE) of $50,563.15, and an R^2 Score of 0.6394 (explaining ~64% of total housing price variance), effectively balancing regularization and predictive accuracy.

2. Problem Statement & Technical Requirements

Accurate real estate valuation requires capturing complex geographical, demographic, and structural attributes. To fulfill Task Two requirements, the following methodology was executed:

- **1. Ingestion & Preprocessing:** Load the California Housing dataset (20,640 records), convert and inspect data types, handle missing total_bedrooms values using SimpleImputer (median), and map the target variable median_house_value.
- **2. Exploratory Data Analysis:** Perform correlation matrix analysis and construct correlation heatmaps to assess linear feature relationships with housing values.
- **3. Feature Engineering & ColumnTransformer:** Separate numerical features (scaling with StandardScaler) and categorical location features (ocean_proximity transformed via OneHotEncoder).
- **4. Regularized Model Exploration:** Train and evaluate linear regression models, including Lasso (L1 regularization) and Ridge (L2 regularization).
- **5. Hyperparameter Tuning & Evaluation:** Execute 5-Fold GridSearchCV to determine optimal regularization strengths (alpha) and evaluate generalization via Residual Plots and Error Histograms.

3. Dataset Analysis & Pipeline Architecture

3.1 Exploratory Data Analysis & Imputation

The dataset comprises 20,640 block group observations across 10 attributes. During initial inspection, total_bedrooms was found to contain 207 missing values (1.0% missingness). A median imputation strategy was integrated within scikit-learn's pipeline to preserve data integrity and prevent data leakage.

Correlation heatmap analysis identified median_income as the single strongest linear predictor of median_house_value ($r \approx +0.688$). Geographic coordinates (latitude and longitude) exhibited a strong negative mutual correlation (-0.92) reflecting California's diagonal coastline, while structural attributes (total_rooms, total_bedrooms, population, households) displayed high collinearity ($r > 0.85$).

3.2 Pipeline Architecture

Preprocessing was modularized using ColumnTransformer to apply separate transformation branches:

- **Numerical Branch (8 features):** housing_median_age, total_rooms, total_bedrooms, population, households, median_income, longitude, latitude transformed using SimpleImputer(strategy='median') and StandardScaler().
- **Categorical Branch (1 feature):** ocean_proximity ('NEAR BAY', '<1H OCEAN', 'INLAND', 'NEAR OCEAN', 'ISLAND') encoded via OneHotEncoder(drop='first', sparse_output=False).

4. Model Development & Hyperparameter Optimization

To prevent overfitting and handle feature multicollinearity, regularized regression models (Lasso and Ridge) were trained and optimized using 5-Fold GridSearchCV across a logarithmic hyperparameter grid for alpha: [0.001, 0.01, 0.1, 1, 10, 100, 1000].

4.1 Lasso Regression (L1 Regularization)

Lasso imposes an L1 penalty on feature coefficients, encouraging feature sparsity. GridSearchCV selected alpha = 100 as the optimal regularization parameter (increasing max_iter=10000 to ensure convergence).

4.2 Ridge Regression (L2 Regularization)

Ridge imposes an L2 penalty, shrinking feature coefficients smoothly without removing variables. GridSearchCV identified alpha = 10 as the optimal penalty parameter, delivering stable coefficient estimates across collinear housing features.

5. Model Evaluation & Comparison

Models were evaluated on a hold-out test set (20% split, 4,128 samples). Predictive performance was quantified using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2):

Model Pipeline	Optimal Hyperparameter	Test RMSE ($)	Test MAE ($)	Test R^2 Score
Lasso Regression	alpha = 100.0	$69,172.45	$50,582.10	0.6392
Ridge Regression	alpha = 10.0	$69,150.31	$50,563.15	**0.6394** ☑

6. Model Diagnostics & Key Valuation Insights

Model diagnostic plots (Residual Analysis and Predicted vs. Actual distributions) revealed several important physical and statistical observations:

- **1. Dominance of Median Income:** median_income had the highest positive standardized coefficient, confirming that local purchasing power is the primary valuation driver in California housing markets.
- **2. Ocean Proximity Impact:** One-hot encoded categories confirmed that proximity to the coast ('INLAND' vs. '<1H OCEAN' / 'NEAR BAY') provides substantial baseline price premiums.
- **3. Price Capping Artifacts:** Scatter plots of Predicted vs. Actual values showed a sharp horizontal boundary at $500,001 due to target truncation in the original census collection. Tree-based non-linear estimators or target transformation strategies are recommended as future steps to better model high-end properties.

7. Conclusions & Recommendations

- **Model Deployment:** Ridge Regression with alpha = 10 is selected as the primary linear baseline due to superior test R^2 (0.6394) and robust handling of collinear physical attributes.
- **Production Packaging:** The ColumnTransformer and fitted Ridge estimator should be exported as a single joblib pipeline artifact for direct inference in valuation API services.